Classical versus Quantum Machine Learning for Credit Default Prediction: An Honest Benchmark at Matched Sample Sizes

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Abstract

We benchmark three classical credit-scoring models against two quantum machine-learning models — a fidelity quantum-kernel support vector classifier (QK-SVC) and a variational quantum classifier (VQC) — on the UCI "Default of Credit Card Clients" dataset (30,000 real Taiwanese credit-card accounts). The evaluation is finance-aware: beyond AUC, PR-AUC and calibration, we score every model under an asymmetric expected-loss matrix and NT$ profit curves over the approval threshold. Because fidelity kernels cost $O(n^2)$ circuit evaluations, quantum models can only be trained on small subsamples; our central methodological rule is that *classical baselines are re-trained on exactly the same subsamples* ($n \in \{500, 1000, 2000\}$, three independent draws), so no comparison ever crosses sample sizes. At every matched size the best classical model matches or exceeds both quantum models (at $n = 2000$: best classical test AUC 0.7691 vs. best quantum 0.7007, a gap of 0.0684), while training orders of magnitude faster. **No quantum advantage is claimed — or observed — in this regime.** The contribution is the protocol and the measured operating envelope: an honest, reproducible baseline that future quantum-hardware claims on tabular credit risk can be checked against. All numbers regenerate from one command in about 7 minutes on a laptop.

1 Introduction

Credit scoring is one of the oldest industrial applications of statistical learning [7, 9], and one of the most heavily benchmarked [8]. It is also, lately, a popular showcase for quantum machine learning (QML), where quantum feature maps promise classically intractable kernels [3, 4]. Provable quantum speedups for supervised learning exist only for contrived problems [5]; whether *any* benefit survives contact with a real, noisy, tabular credit dataset at the sample sizes quantum methods can actually process is an empirical question — and one that is easy to answer misleadingly. A recurring failure mode in applied QML papers is the uneven comparison: a quantum model trained on hundreds of points is compared against a classical model trained on the full dataset (or vice versa), with accuracy as the only metric and wall-clock cost unreported.

This working paper does the boring version of the experiment, deliberately. Our contributions are methodological: (i) a *matched-subsample fairness rule* — every classical baseline is re-trained on the exact stratified subsamples the quantum models use; (ii) a *finance-aware evaluation* — an asymmetric expected-loss matrix with its Bayes-optimal operating threshold, and portfolio profit curves in NT$ using each client's actual credit limit as exposure; (iii) *honest cost accounting* — wall-clock time for every phase of every model, including quantum state preparation; and (iv) full reproducibility — fixed seeds, defensive data validation, known-answer unit tests (kernel

symmetry/PSD, cost-matrix algebra), and a single command that regenerates every number and figure in this paper.

2 Data

We use the UCI "Default of Credit Card Clients" dataset (id 350) [1, 2]: 30,000 Taiwanese credit-card clients observed in October 2005, 23 raw features (credit limit, demographics, six months of repayment status `PAY_0, PAY_2-PAY_6`, bill and payment amounts), and a binary target — default on the next monthly payment. The default rate is 22.12% (6,636 of 30,000); classes are imbalanced but not extremely so. The raw `.xls` is fetched once and cached (SHA-256 recorded in `results/data_manifest.json`); a defensive parser validates the documented schema (exact row count, value ranges, no missing values) before any modelling. Cleaning is minimal and fully documented in `docs/DATA_NOTES.md`: undocumented `EDUCATION` codes $\{0, 5, 6\}$ (345 cells) and `MARRIAGE` code 0 (54 cells) are collapsed into their "others" categories; no rows are dropped; no values are imputed. We hold out a fixed stratified 25% test set ($n_{\text{test}} = 7,500$, seed 42) used by *every* model in this paper; all tuning happens inside the training set.

3 Methodology

3.1 Classical models and protocol

Three standard baselines are trained on the full training set ($n = 22,500$): ℓ_2 logistic regression with balanced class weights and isotonic probability calibration; a random forest (400 trees, balanced subsample weights); and histogram gradient boosting (HGB) with balanced class weights. Categorical variables are one-hot encoded and numerics standardised for the linear model inside the pipeline (no leakage); tree ensembles consume the raw integer-coded features. We report 5-fold stratified cross-validation means ± standard deviations, held-out test metrics, and quantile-binned reliability diagrams. Operating thresholds are selected on out-of-fold training predictions only, then applied unchanged to the test set.

3.2 Finance-aware, cost-sensitive evaluation

Accuracy is the wrong loss for lending. Let $y_i = 1$ denote default and let a model reject applicant i when its predicted default probability p_i exceeds a threshold t. With unit costs c_{FN} (approving a future defaulter) and c_{FP} (rejecting a good client), the empirical expected cost per applicant is

$$\widehat{C}(t) \;=\; \frac{1}{n}\Big(c_{\text{FN}}\,\text{FN}(t) \;+\; c_{\text{FP}}\,\text{FP}(t)\Big), \tag{1}$$

and for perfectly calibrated probabilities the Bayes-optimal rule rejects iff $p \geq t^*$ with

$$t^* \;=\; \frac{c_{\text{FP}}}{c_{\text{FP}} + c_{\text{FN}}} \tag{2}$$

[6]. We adopt the classic credit-scoring asymmetry $c_{\text{FN}} : c_{\text{FP}} = 5 : 1$, giving $t^* = 1/6 \approx 0.167$; the distance between a model's empirical cost-minimising threshold and t^* is itself a calibration diagnostic. Complementarily, we compute portfolio profit with heterogeneous exposures, approving iff $p_i < t$:

$$\Pi(t) \;=\; \sum_{i\,:\,p_i<t} \Big[(1-y_i)\,m\,L_i \;-\; y_i\,\lambda\,L_i\Big], \tag{3}$$

where L_i is client i's actual credit limit (`LIMIT_BAL`, in NT$), $m = 0.10$ a stylised revenue margin on performing accounts and $\lambda = 0.60$ a stylised loss-given-default fraction. These economic parameters are illustrative and clearly flagged as such; the full curves over t (Figure 2) expose the sensitivity.

3.3 Fidelity quantum kernel

Following Havlíček et al. [3], data are embedded by the second-order Pauli-Z feature map $\mathcal{U}_{\Phi(x)}$ (`ZZFeatureMap`, 2 repetitions): after Hadamards, single- and two-qubit phase rotations with angles $\phi_i(x) = x_i$ and $\phi_{ij}(x) = (\pi - x_i)(\pi - x_j)$. The kernel is the state fidelity

$$K(x, x') \;=\; \big|\langle \Phi(x) \,|\, \Phi(x')\rangle\big|^2, \tag{4}$$

consumed by a standard SVC with balanced class weights (precomputed Gram matrix; C chosen by inner 3-fold CV over the same grid as the classical RBF-SVC, its exact classical twin). On real hardware K is estimated by the compute–uncompute circuit $\mathcal{U}^\dagger_{\Phi(x')}\mathcal{U}_{\Phi(x)}|0\rangle^{\otimes d}$ (probability of the all-zeros outcome); on the statevector simulator used here Eq. (4) is evaluated exactly, i.e. our quantum results are *noise-free upper bounds* on NISQ performance. Since features must fit d qubits, we select the top-d features by mutual information with the label ($d = 4$ primary, $d = 8$ robustness), estimated on each training subsample only, then min–max scale them to angles in $[0, \pi]$ (test values clipped to the training range). MI selection is preferred over PCA to keep the encoded features interpretable and to avoid rotating integer-coded categoricals into dense components; PCA is implemented as an alternative.

3.4 Variational quantum classifier

The VQC applies a RealAmplitudes ansatz $U(\theta)$ (3 entangling repetitions, 16 parameters at $d = 4$) to the same feature-map states and measures the qubit-parity observable $P = Z^{\otimes d}$:

$$p(x;\theta) \;=\; \tfrac{1}{2}\Big(1 - \langle \Phi(x)|U(\theta)^\dagger P\, U(\theta)|\Phi(x)\rangle\Big), \tag{5}$$

trained by COBYLA (250 iterations, 2 restarts, best kept) on class-weighted cross-entropy. Because P is diagonal and the feature-map states are cached, each objective evaluation reduces to applying one $2^d \times 2^d$ unitary to the cached state matrix — exact simulation, reported timings included.

3.5 The fairness rule

Building the Gram matrix costs $O(n^2)$ kernel evaluations ($\sim$2M circuits at $n = 2000$ on hardware), so quantum models are trained on stratified subsamples $n \in \{500, 1000, 2000\}$ with three independent draws (seeds 0/1/2). **Every classical baseline is re-trained on exactly the same subsamples.** All models — classical and quantum, every size — are evaluated on the same fixed 7,500-client test set, and decision costs are compared at the same fixed Bayes threshold $t^* = 1/6$ (no per-model threshold tuning at small n). We never compare quantum-on-2000 with classical-on-22,500.

4 Results

4.1 Part 1 — classical baselines on the full training set

Table 1 summarises the full-data study. Gradient boosting (HGB) performs best (test AUC 0.7774, PR-AUC 0.5528, Brier 0.1792). Reliability diagrams (Figure 1) show the isotonic-calibrated logistic model closest to the diagonal, consistent with its empirical cost-optimal threshold landing nearest the Bayes value $t^* = 1/6$ (Eq. 2): Logistic regression (calibrated) $\hat{t} = 0.162$; Random forest $\hat{t} = 0.282$; Gradient boosting (HGB) $\hat{t} = 0.404$. Cost and profit curves are shown in Figure 2; at its out-of-fold profit-optimal threshold, Gradient boosting (HGB) attains NT\$34.2M on the test portfolio at an approval rate of 45.3%.

Table 1: Full-training-set classical baselines ($n = 22,500$; test $n = 7,500$). CV: 5-fold stratified mean ± std. Cost per applicant (Eq. 1) at the OOF-selected threshold; profit (Eq. 3) in NT$ millions.

Model	CV AUC	Test AUC	PR-AUC	Brier	Cost@t^*_{OOF}	Profit (M)	Fit (s
Logistic regression (calibrated)	0.7281 ± 0.0072	0.7189	0.4966	0.1424	0.6376	17.6	1.9
Random forest	0.7795 ± 0.0046	0.7721	0.5527	0.1495	0.5736	30.8	3.7
Gradient boosting (HGB)	0.7822 ± 0.0059	0.7774	0.5528	0.1792	0.5612	34.2	0.1

Table 2: Matched-subsample test AUC (mean ± std over 3 identical stratified draws; C = classical, Q = quantum). Same subsamples, same test set, same threshold rule for every model.

Model	$n = 500$	$n = 1000$	$n = 2000$	Time@2000 (s)
Logistic regression (calibrated) (C)	0.6949 ± 0.0267	0.7118 ± 0.0033	0.7148 ± 0.0017	0.0
Random forest (C)	0.7513 ± 0.0060	0.7613 ± 0.0013	0.7691 ± 0.0042	0.6
Gradient boosting (HGB) (C)	0.7038 ± 0.0085	0.7158 ± 0.0093	0.7352 ± 0.0114	0.3
SVC, RBF kernel (classical twin) (C)	0.7193 ± 0.0070	0.7368 ± 0.0038	0.7486 ± 0.0051	1.5
QK-SVC, fidelity kernel, d=4 (Q)	0.6875 ± 0.0026	0.7011 ± 0.0119	0.6891 ± 0.0247	7.2
QK-SVC, fidelity kernel, d=8 (Q)	0.6643 ± 0.0173	0.6812 ± 0.0137	0.7007 ± 0.0029	28.4
VQC, d=4 (Q)	0.6936 ± 0.0040	0.6811 ± 0.0189	0.6713 ± 0.0210	7.1

What drives default risk. Permutation importance on the test set (Figure 3) is dominated by the most recent repayment status `PAY_0`, followed by `LIMIT_BAL`, `BILL_AMT1` — recent delinquency, not demographics, carries the signal, in line with the credit-scoring literature [1, 8]. Top-5 AUC drops: `PAY_0` (0.083), `LIMIT_BAL` (0.018), `BILL_AMT1` (0.016), `PAY_AMT3` (0.010), `PAY_AMT1` (0.007).

4.2 Part 2 — matched subsamples: the honest comparison

Table 2 is the central result. At every matched training size the best classical model is at least as good as the best quantum model; at $n = 2000$ the gap between the best classical (Random forest, 0.7691) and the best quantum (QK-SVC, fidelity kernel, d=8, 0.7007) is 0.0684 AUC. The quantum kernel at $d = 8$ trails $d = 4$ at the two smaller sizes and edges past it only at $n = 2000$, remaining well below every classical baseline; the VQC trails throughout, as expected for a 16-parameter model. In expected-cost terms (Table 3) the classical RBF-SVC is the cheapest decision-maker at every size; the Platt-calibrated quantum-kernel probabilities are competitive with the tree ensembles at this fixed threshold — a calibration effect, not a ranking reversal. The VQC's parity-derived probabilities concentrate near 0.5, so at the fixed 1/6 threshold it rejects essentially the whole portfolio: its expected cost sits at the reject-all bound. Figure 4 overlays learning curves; Figure 5 shows honest wall-clock costs — the QK-SVC pays a large simulation overhead (dominated by test-set embedding and the $n_{\text{test}} \times n$ kernel block), and on hardware the same object would cost millions of shots. Figure 6 visualises a Gram sub-block: within-class and between-class fidelities are barely distinguishable, a first hint of why the kernel does not separate this data better than its classical twin.

4.3 Hardware estimation path

The repository implements this estimator behind an explicit `-hardware` flag (`experiments/run_hardware.py`): it reads the user's saved `QiskitRuntimeService` account from `~/.qiskit` (tokens never enter the repository), submits one job of at most 28 compute–uncompute circuits to the least-busy backend, and reports measured versus simulated kernel entries. It was *not* executed for this draft: the build environment holds no QPU credentials, and the tables above are exact-simulation results by design. Hardware numbers, when produced, land in `results/hardware_kernel.json`

Table 3: Expected cost per applicant (Eq. 1) at the fixed Bayes threshold $t^* = 1/6$; lower is better.

Model	$n = 500$	$n = 1000$	$n = 2000$
Logistic regression (calibrated)	0.6862 ± 0.0286	0.6960 ± 0.0102	0.6871 ± 0.0100
Random forest	0.6685 ± 0.0115	0.6516 ± 0.0090	0.6389 ± 0.0204
Gradient boosting (HGB)	0.7300 ± 0.0254	0.6948 ± 0.0120	0.6386 ± 0.0176
SVC, RBF kernel (classical twin)	0.6402 ± 0.0117	0.6251 ± 0.0107	0.6007 ± 0.0151
QK-SVC, fidelity kernel, d=4	0.6670 ± 0.0193	0.6321 ± 0.0081	0.6288 ± 0.0150
QK-SVC, fidelity kernel, d=8	0.7060 ± 0.0327	0.6732 ± 0.0067	0.6574 ± 0.0062
VQC, d=4	0.7788 ± 0.0000	0.7787 ± 0.0001	0.7796 ± 0.0015

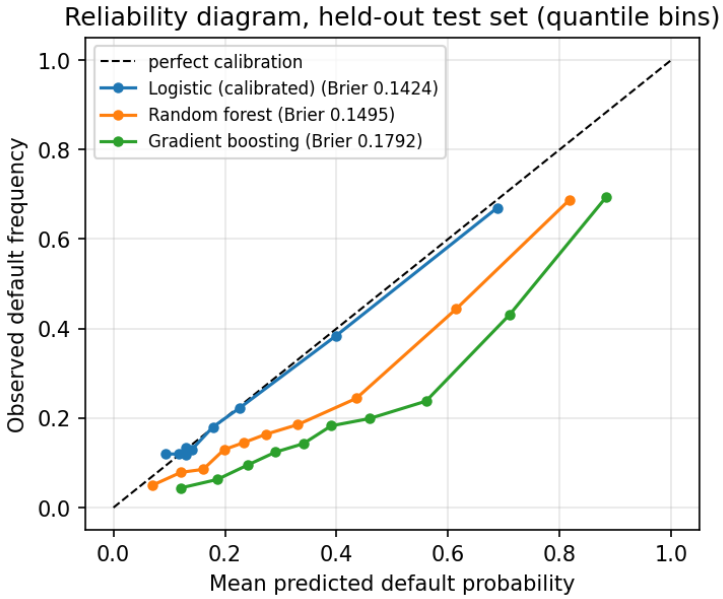


Figure 1: Reliability diagrams on the held-out test set (quantile bins).

and are reported verbatim.

5 Limitations

This is a single dataset, from 2005, from one market; conclusions are about the *measured regime*, not about QML in general. The economic parameters ($c_{\mathrm{FN}} : c_{\mathrm{FP}}, m, \lambda$) are stylised; profit levels shift with them even though model rankings here do not. Quantum models were simulated exactly (no shot noise, no device noise): real-hardware results would be strictly worse, which only strengthens the negative finding, but we did not measure the degradation beyond providing the estimator. Feature compression to $d \in \{4, 8\}$ discards information by construction; the classical models in the matched study consume all 23 features, which is deliberate — the constraint *is* the point — but a reader may also compare the RBF-SVC row as a same-feature-count proxy. Hyperparameter search was intentionally light and symmetric (C grid only); three subsample seeds bound variability but do not support formal significance tests. Finally, the VQC's optimisation landscape is nonconvex; different ansätze or optimisers could close part of its gap, though hardly the kernel's.

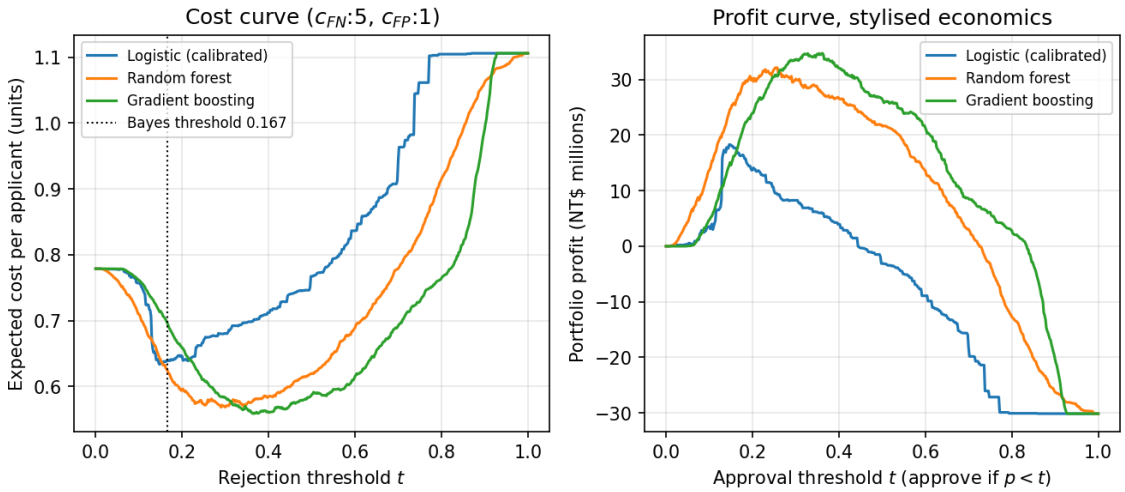


Figure 2: Left: expected cost per applicant vs. rejection threshold, with the Bayes threshold $t^* = 1/6$. Right: NT$ portfolio profit vs. approval threshold under the stylised economics of Eq. (3).

6 Conclusion

On real credit-risk data, with a fair protocol and finance-aware metrics, classical machine learning wins at every sample size a fidelity quantum kernel can currently reach, and it wins while running orders of magnitude faster. **No quantum advantage is claimed, and none was observed.** The value of this study is the ruler, not the record: a reproducible, cost-honest, matched-sample benchmark that future quantum claims on tabular financial data can be measured against. The full pipeline regenerates in about 7 minutes on a laptop from `experiments/run_all.py`.

Reproducibility and disclosure

Code, tests and this paper's sources are in the `quantum-credit-mqfs` repository (MIT licence). Environment: Python 3.10.12, scikit-learn 1.7.2, Qiskit 2.5.0. Built with AI-assisted engineering; every design decision, formula and validation was reviewed and is owned by the author.

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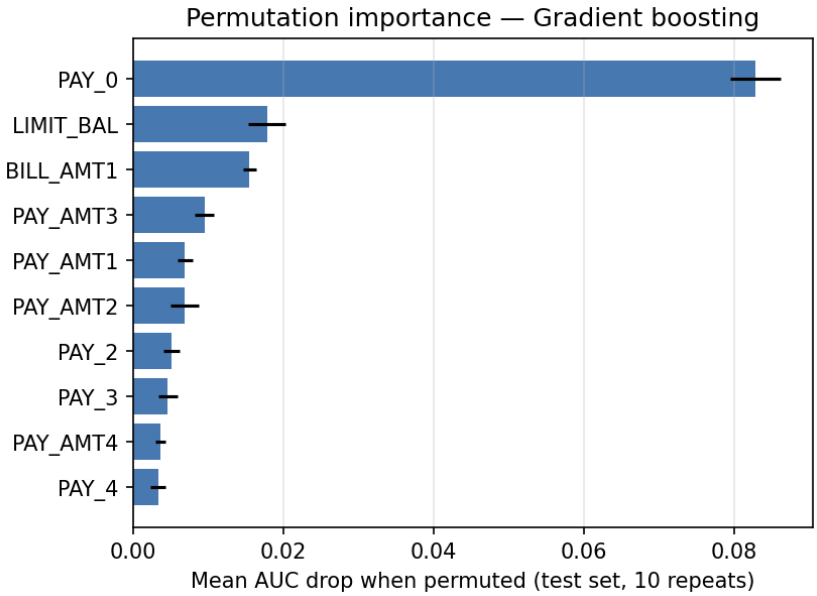


Figure 3: Permutation feature importance (test set, 10 repeats) for the best full-data model.

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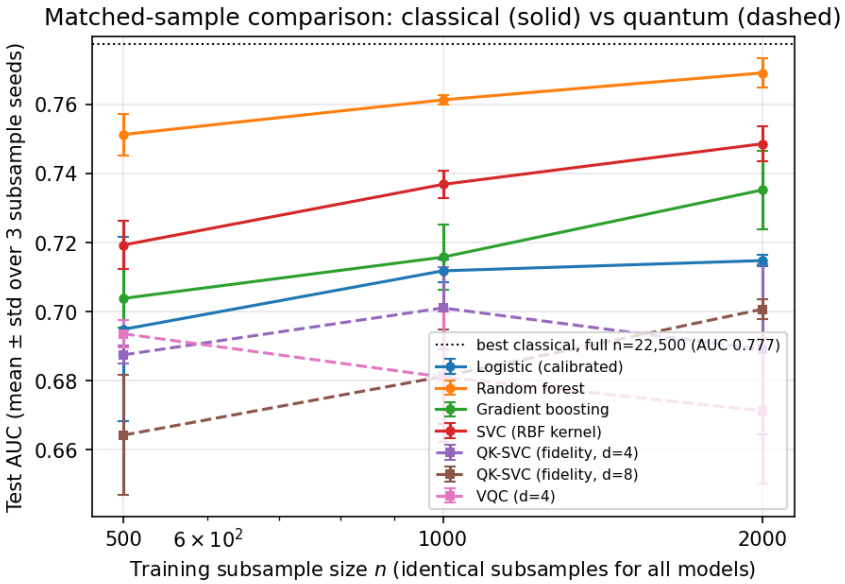


Figure 4: Matched-sample learning curves: classical (solid) vs. quantum (dashed), mean ± std over three identical subsample draws.

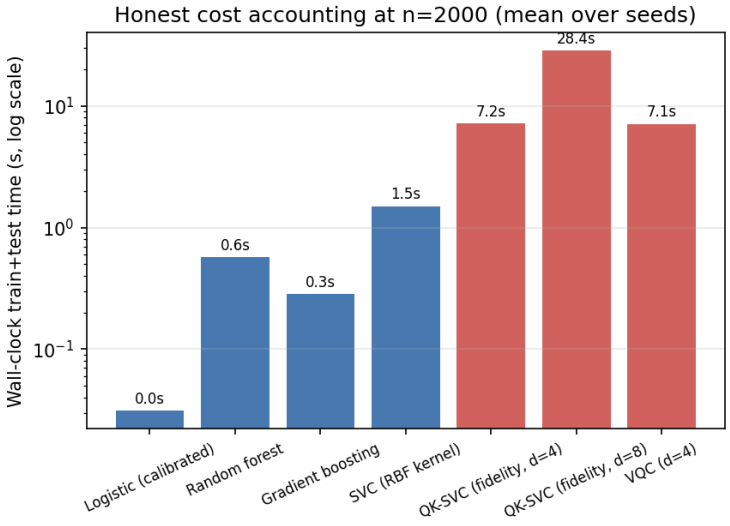


Figure 5: Honest wall-clock accounting at $n = 2000$ (log scale). Quantum timings are exact-statevector simulation; hardware execution would add shot and queue overheads.

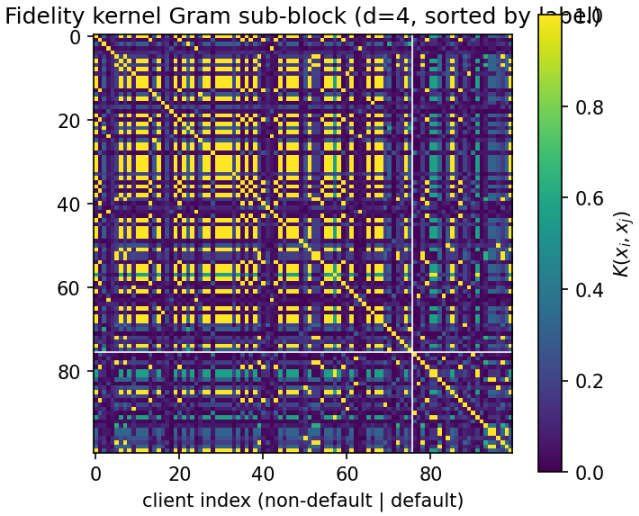


Figure 6: Fidelity-kernel Gram sub-block ($d = 4$, 100 training points sorted by label).